

WEATHER TREND FORECASTING

Global Weather Repository dataset analysis

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Github: <https://github.com/rajaraman04/Weather-Forecasting/tree/main>

ABSTRACT

This project analyzes the Global Weather Repository dataset to identify temporal, geographical, environmental, and air-quality patterns and to forecast future temperature trends. The analysis followed a complete data-science workflow, including data inspection, preprocessing, exploratory data analysis, anomaly detection, spatial comparison, feature engineering, model development, and evaluation. The `last_updated` field was used as the primary time variable, while missing values, duplicates, invalid measurements, and statistical outliers were examined during preprocessing. Temperature outliers were retained because they could represent genuine extreme-weather conditions.

Exploratory analysis revealed noticeable differences in temperature, precipitation, and air quality across locations, countries, continents, and latitude zones. Isolation Forest was applied to identify unusual combinations of weather conditions. For forecasting, Bujumbura, Burundi, was selected because it contained the longest suitable daily temperature history. After interpolating three missing calendar dates and creating lag, rolling, and calendar features, 783 observations were used for model development.

Naive Persistence, Ridge Regression, Random Forest, SARIMA, and two ensemble methods were evaluated using a chronological train-validation-test split. SARIMA achieved the best test performance with an MAE of 1.145°C, RMSE of 1.521°C, R^2 of 0.083, and sMAPE of 5.097%, reducing MAE by approximately 20.8% compared with the naive baseline. Feature-importance analysis showed that recent temperature history was more informative than calendar features. The final SARIMA model produced a seven-day forecast ranging from approximately 20.94°C to 21.18°C.

Project Objective

The objective of this project is to analyse the Global Weather Repository dataset and develop a forecasting workflow for future temperature trends. This project examines historical weather observations from cities around the world to identify temporal, geographical, environmental, and seasonal patterns. It also evaluates multiple statistical and machine-learning forecasting approaches to determine which method provides the most reliable temperature predictions for the selected location.

The analysis was designed to answer the following questions:

- What major temperature and precipitation patterns are present in the dataset?
- How do weather conditions vary across countries, continents, and latitude regions?
- Are there unusual or anomalous weather observations?
- How are weather variables associated with air-quality measurements?
- Which historical and calendar features are most useful for predicting temperature?
- Which forecasting model performs best on unseen data?
- What temperatures are expected during the next seven-day period?

Assessment Scope:

Basic Assessment includes:

- Inspecting and cleaning the dataset.
- Handling missing values, duplicate records, invalid values and outliers.
- Normalizing model features wherever required.
- Performing Exploratory Data Analysis (EDA)
- Identifying trends, correlations and weather patterns
- Creating temperature and precipitation visualizations
- Building a basic forecasting model
- Perform time-series analysis using the last_update field

Advanced Assessment includes:

- Multivariable anomaly detection
- Comparison of multiple forecasting models
- Ensemble forecasting
- Seasonal and climate pattern analysis
- Air-quality and environmental correlation analysis
- Feature importance analysis using multiple techniques
- Spatial and geographical visualization
- Country and continent level weather comparisons
- Generation of seven day future temperature forecast

Project Workflow

The project followed the workflow below:

1. Load and inspect the weather dataset.
2. Validate data quality and prepare datetime information.
3. Analyze temperature, precipitation, weather, and air-quality patterns.
4. Detect unusual observations (anomaly) using statistical and machine-learning methods.
5. Compare weather conditions across geographical regions.
6. Select a suitable city for time-series forecasting.
7. Create lag, rolling, and calendar features.
8. Split the data chronologically into training, validation, and test periods.
9. Train and evaluate multiple forecasting models.
10. Create ensemble forecasts.
11. Analyze feature importance.
12. Retrain the best forecasting model using all available historical data.
13. Generate a seven day future temperature forecast.
14. Summarize findings, limitations and future improvements.

2. DATASET OVERVIEW

Dataset Source:

The analysis used the Global Weather Repository dataset available through Kaggle. The dataset contains daily weather observations for cities across multiple countries and continents.

It combines meteorological, geographical, astronomical, and air-quality information, making it suitable for exploratory analysis, anomaly detection, spatial comparison, and time-series forecasting.

Source: Kaggle - GlobalWeatherRepository.csv

Forecasting target: temperature_celsius

Time variable used: last_updated

Dataset Dimensions:

Number of observations (rows)	154556
Number of variables (columns)	41
Number of countries	211
Number of locations	286
Earliest observation date	May 16, 2024

Latest observation date	July 21, 2026
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Each row represents a weather observation for a particular location and date. The dataset includes more than 40 variables describing weather conditions, geographical information, air quality, and astronomical measurements.

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	condition_text
0	Afghanistan	Kabul	34.520	69.180	Asia/Kabul	1715849100	2024-05-16 13:15	26.600	79.800	Partly Cloudy
1	Albania	Tirana	41.330	19.820	Europe/Tirane	1715849100	2024-05-16 10:45	19.000	66.200	Partly cloudy
2	Algeria	Algiers	36.760	3.050	Africa/Algiers	1715849100	2024-05-16 09:45	23.000	73.400	Sunny
3	Andorra	Andorra La Vella	42.500	1.520	Europe/Andorra	1715849100	2024-05-16 10:45	6.300	43.300	Light drizzle
4	Angola	Luanda	-8.840	13.230	Africa/Luanda	1715849100	2024-05-16 09:45	26.000	78.800	Partly cloudy

Main Feature Categories

The dataset variables were grouped into the following categories.

Location and Time Information - These variables identify where and when each weather observation was recorded.

Examples: country, location_name, latitude, longitude, timezone, last_updated

Temperature Measurements - These variables describe the recorded and perceived temperature conditions.

Examples: temperature_celsius, temperature_fahrenheit, feels_like_celsius, wind_chill_celsius, feels_like_fahrenheit, heat_index_celsius

The main forecasting target - temperature_celsius

Precipitation and Atmospheric Conditions - These variables describe rainfall and general atmospheric conditions.

Examples: precip_mm, precip_in, humidity, pressure_mb, pressure_in, cloud, visibility_km

Wind Measurements - Wind related features include: wind_kph, wind_mph, wind_degree, wind_direction, gust_kph

These variables help describe the movement and intensity of local atmospheric conditions.

Air quality Measurements - The dataset also contains pollutant measurements such as Carbon monoxide, Ozone, Nitrogen dioxide, Sulphur dioxide, PM2.5, PM10, US Environmental Protection Agency air-quality index, Great Britain air-quality index. These variables were used to examine relationships between environmental conditions and weather patterns.

Astronomical Measurements - This includes Sunrise, Sunset, Moonrise, Moonset, Moon phase, Moon illumination. These features provide additional context about daylight and seasonal conditions.

Time Variable and Forecasting Target

The *last_updated* field was selected as the primary time variable because the assessment specifically required it to be used for time-series analysis.

The field was converted from text into a datetime format and used to

- Sort observations chronologically
- Extract calendar features
- Analyze monthly and seasonal patterns
- Create a continuous daily temperature series
- Divide the forecasting data into training, validation, and test periods

The primary forecasting target was *temperature_celsius*. Temperature in celsius was selected because it is directly interpretable and consistently available across the selected historical series.

Forecasting Location

Although the dataset contains observations from cities around the world, forecasting was performed at the individual-location level. Combining all cities into one time series would mix different climates, geographical regions, and seasonal cycles. Therefore, locations were compared based on the amount and continuity of their historical data.

Bujumbura, Burundi was selected because it contained the longest suitable daily temperature history for forecasting.

3. DATA PREPARATION

Datetime Processing - The *last_updated* column was converted to datetime format and used as the primary time variable. Observations were sorted chronologically before trend analysis and forecasting.

Additional calendar features were created from this field, including Month, Day of year, Day of week, Quarter, Season.

Missing Values and Duplicates

The dataset was checked for missing values and duplicate records. No missing values were found in the original dataset. Duplicate records were also checked before analysis to avoid

For the selected forecasting location, Bujumbura, Burundi the data comprises of: 794 recorded dates, 3 missing calendar dates, 797 dates after creating a continuous daily series.

The missing calendar dates were filled using time-based interpolation so that lag and rolling features could be calculated without gaps.

Invalid-Value Validation

Numerical weather variables were checked for values outside reasonable ranges, including: Humidity outside 0%–100%, Negative precipitation, Negative wind speed, Invalid geographical coordinates, Implausible visibility and pressure values. Values identified as invalid during validation were treated as missing and replaced using median values where necessary.

Outlier Handling

Temperature outliers were identified using the interquartile range method. Observations below: $[Q1 - 1.5 * QR]$ or above: $[Q3 + 1.5 * QR]$ were flagged as statistical outliers.

These observations were retained because unusual temperatures may represent genuine weather conditions rather than data-entry errors. Multivariable anomaly detection was later performed using Isolation Forest.

Feature Engineering

The following forecasting features were created:

Feature	Description
lag_1	Temperature one day earlier
lag_7	Temperature seven days earlier
lag_14	Temperature fourteen days earlier
rolling_mean_7	Average temperature over the previous seven days
rolling_std_7	Temperature variability over the previous seven days
month	Calendar month
day_of_year	Position within the year
day_of_week	Day of the week

Feature Normalization

Ridge Regression requires features to be on comparable scales. Therefore, StandardScaler was applied within a modeling pipeline.

The scaler was fitted only on the training data and then applied to the validation and test data. This prevented information from future observations from leaking into model training.

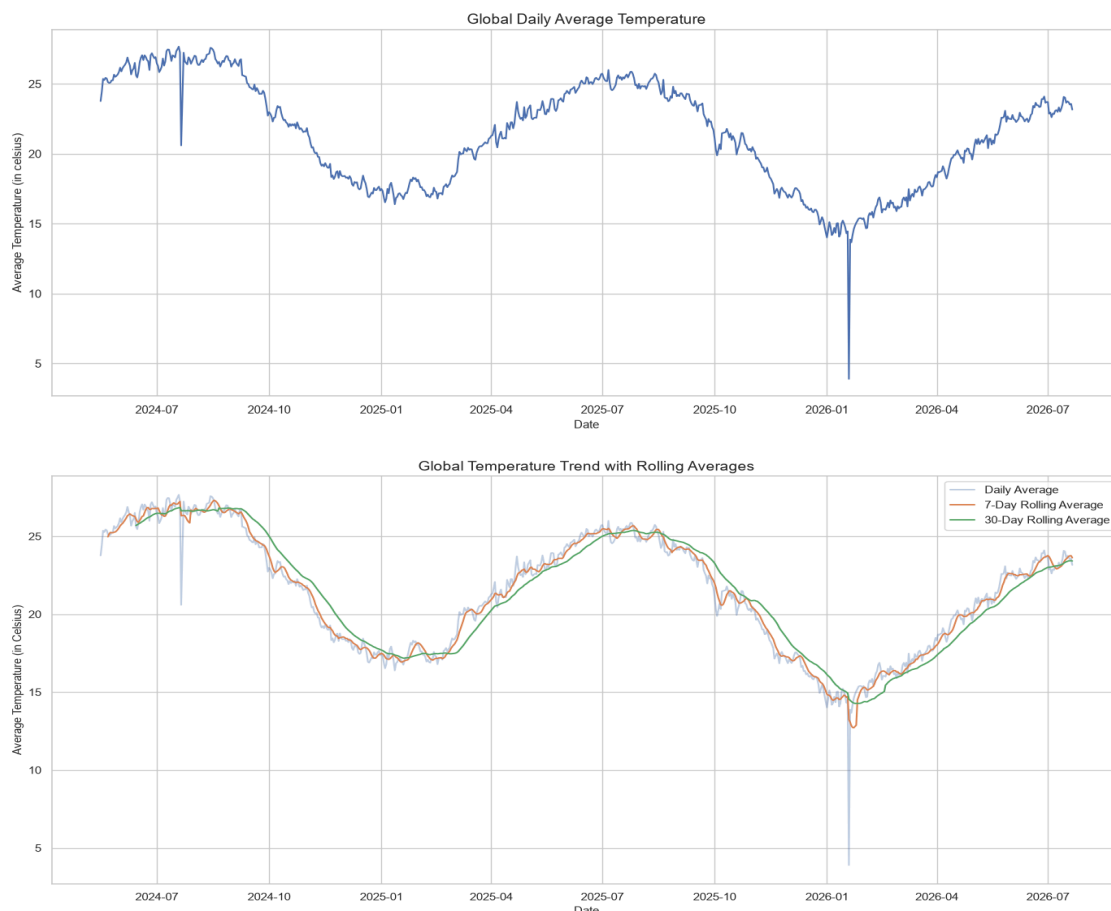
Random Forest did not require normalization because tree-based models are not affected by differences in feature scale. SARIMA modeled the ordered temperature series directly and did not use the engineered feature matrix.

4. EXPLORATORY DATA ANALYSIS

Temperature Analysis

Temperature distributions and time-based trends were analyzed to understand variation across the dataset.

The analysis showed that temperature differed substantially across locations because the dataset contains cities from multiple climate zones. Daily temperature plots also showed short-term fluctuations and broader seasonal movement.



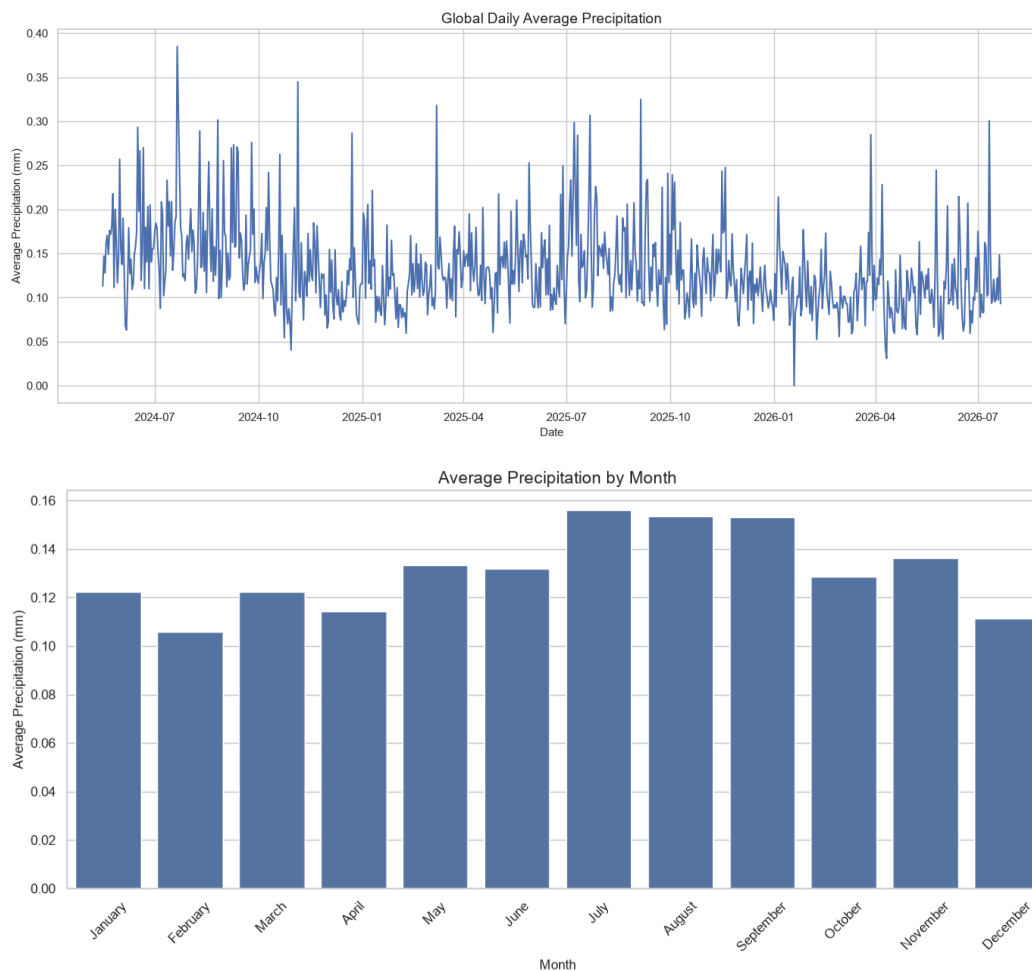
These visualizations demonstrate both the overall spread of temperature values and how temperature changes across dates.

Precipitation Analysis

Precipitation was examined using distribution plots and time-based summaries.

Most observations recorded low or no precipitation, while a smaller number of records contained substantially higher rainfall. This produced a right-skewed precipitation distribution.

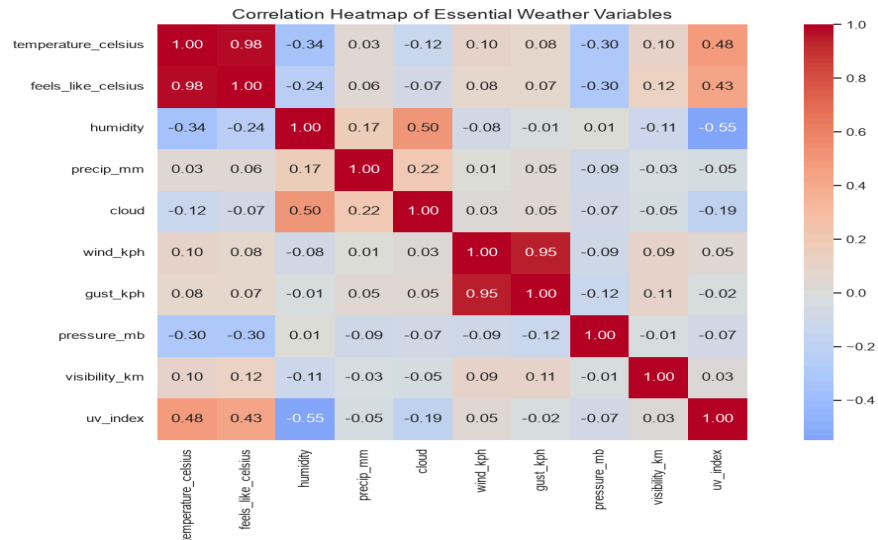
Monthly and location-level comparisons showed that rainfall patterns varied considerably across time and geographical regions.



Correlation Analysis

A correlation matrix was used to examine relationships among major numerical weather variables. The strongest relationships were observed among directly related measurements, such as:

- Temperature and feels-like temperature
- Temperature measured in Celsius and Fahrenheit
- Wind speed measured in kilometres and miles per hour
- Atmospheric pressure measured in millibars and inches
- Precipitation measured in millimetres and inches



Humidity, cloud cover, precipitation, visibility, and temperature also showed varying relationships, although these were weaker and differed across locations.

Temporal and Seasonal Patterns

Calendar features derived from last_updated were used to compare weather conditions by month, season, and day of year.

Temperature and precipitation showed seasonal differences, but the timing and intensity of these patterns varied across locations and hemispheres.

Because the dataset combines cities from different climate regions, global averages may hide important local differences. For this reason, forecasting was performed using one location rather than the full global dataset.

Main EDA Findings

- Temperature varied substantially across cities, countries, and climate regions.
- Precipitation was concentrated in fewer high-rainfall observations.
- Weather variables showed both expected physical correlations and location-specific relationships.
- Seasonal patterns were visible but differed across geographical regions.
- Location-level analysis was necessary because combining all cities into one time series would mix different climates and seasonal cycles.

5. ANOMALY DETECTION

Methodology

Isolation Forest was used to identify unusual weather observations based on multiple variables such as Temperature, Humidity, Precipitation, Atmospheric pressure, Wind speed, Cloud cover, Visibility

This method detects observations that differ strongly from the overall weather pattern without requiring predefined anomaly labels.

Interpretation

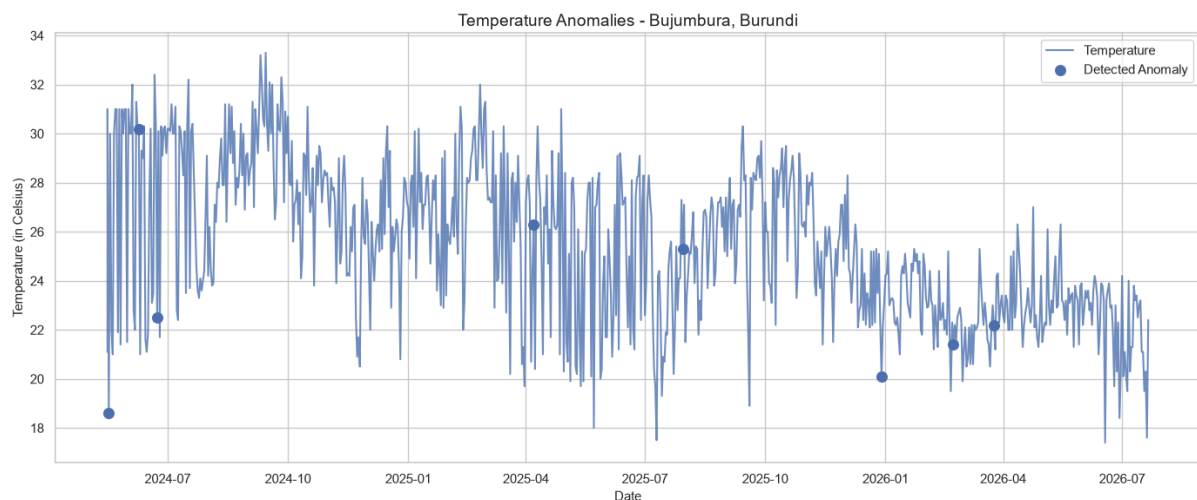
The detected anomalies represented unusual combinations of weather conditions rather than extreme values in only one variable.

Examples include: High temperature with unusual humidity, Heavy precipitation with low visibility, Strong wind combined with abnormal pressure, Unusual cloud-cover and visibility patterns.

The anomalous observations were retained because they could represent genuine extreme-weather conditions, temporary local events, or measurement irregularities.

Main Findings

- Isolation Forest identified a small subset of observations as anomalous.
- Multivariable anomaly detection provided more context than temperature only outlier detection.
- An anomaly does not automatically indicate incorrect data.
- Additional meteorological information would be required to confirm whether each anomaly represents a real weather event or a measurement issue.



6. CLIMATE, GEOGRAPHY AND SPATIAL ANALYSIS

Country and Continent Comparison

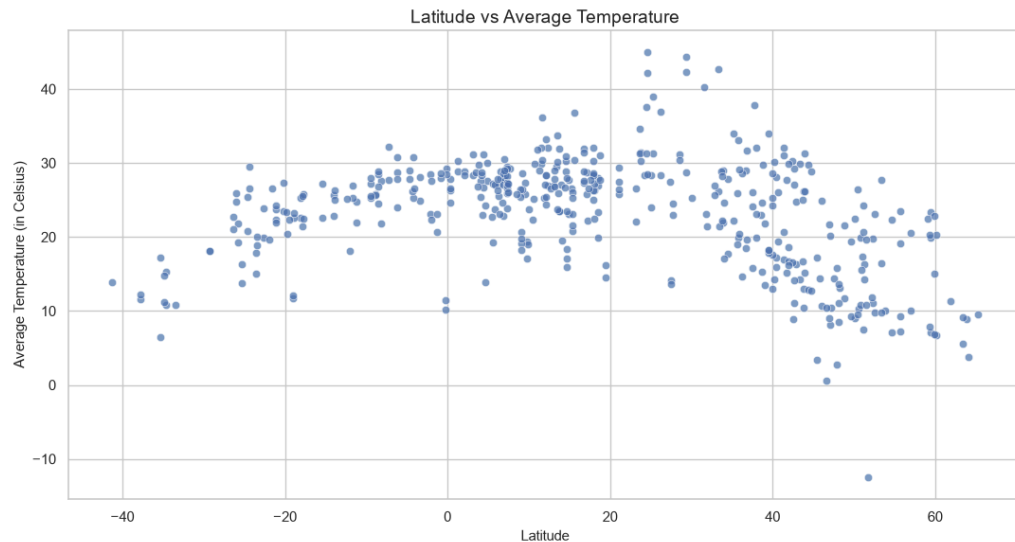
Average temperature and precipitation were compared across countries and continents to identify regional differences.

The analysis showed that weather conditions varied substantially across geographical regions. Tropical and equatorial locations generally recorded warmer temperatures, while locations farther from the equator tended to record lower average temperatures. Precipitation patterns also differed across countries and continents, reflecting differences in regional climate and seasonality.

Latitude and Temperature Relationship

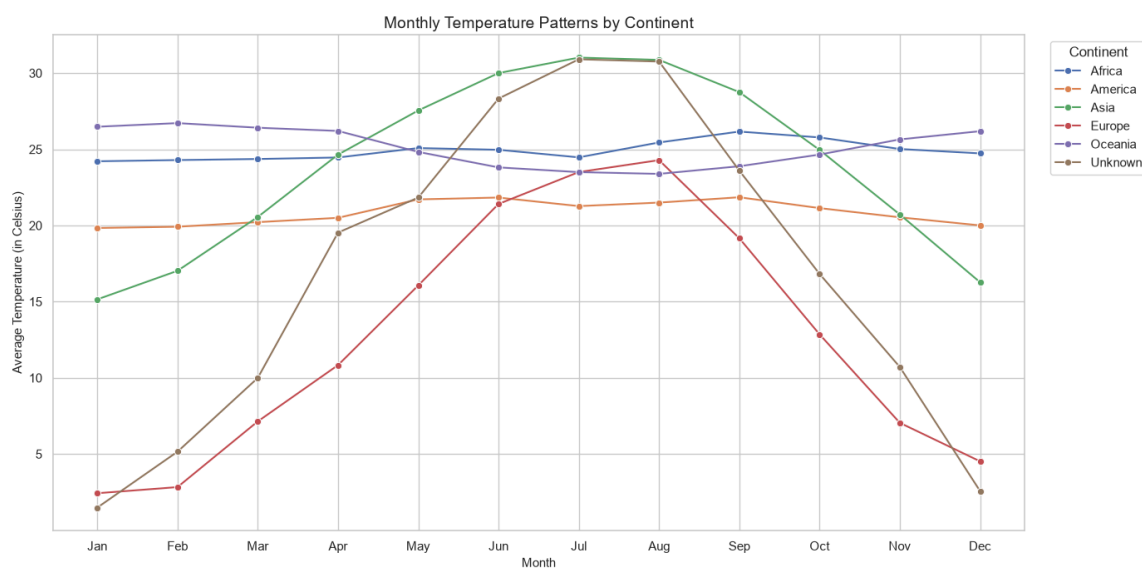
Latitude was compared with temperature to examine how distance from the equator influenced weather conditions.

The results showed a general decline in temperature as absolute latitude increased. However, latitude alone did not fully explain local temperature because other factors such as elevation, coastal proximity, terrain, and seasonal conditions were not included.



Seasonal Climate Patterns

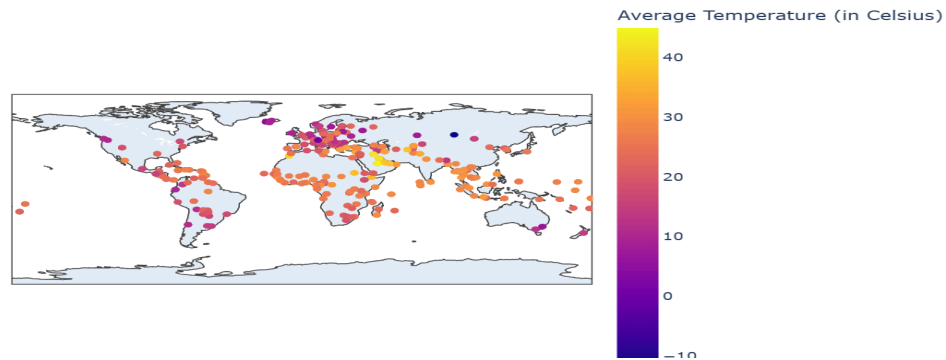
Monthly temperature trends were compared across continents. The timing of warmer and cooler periods differed between regions because cities in the Northern and Southern Hemispheres experience opposite seasonal cycles. These findings show that global averages can hide important regional climate patterns.



Spatial Distribution

Latitude and longitude were used to visualize the geographical distribution of temperature across locations. The spatial analysis showed that temperature was not distributed uniformly. Warmer observations were generally concentrated near lower latitudes, while cooler observations were more common at higher latitudes.

Global Distribution of Average Temperature



Main Findings

- Temperature and precipitation differed clearly across countries and continents.
- Locations closer to the equator generally recorded higher temperatures.
- Seasonal patterns varied across hemispheres and geographical regions.
- Latitude was an important geographical factor but did not fully explain local weather.
- Spatial analysis confirmed that location-level forecasting is more appropriate than combining all cities into one global time series.
- The available dataset supports regional and seasonal comparison but is not long enough to establish long-term climate-change trends.

7. AIR QUALITY AND ENVIRONMENTAL RELATIONSHIPS

Air-Quality Variables

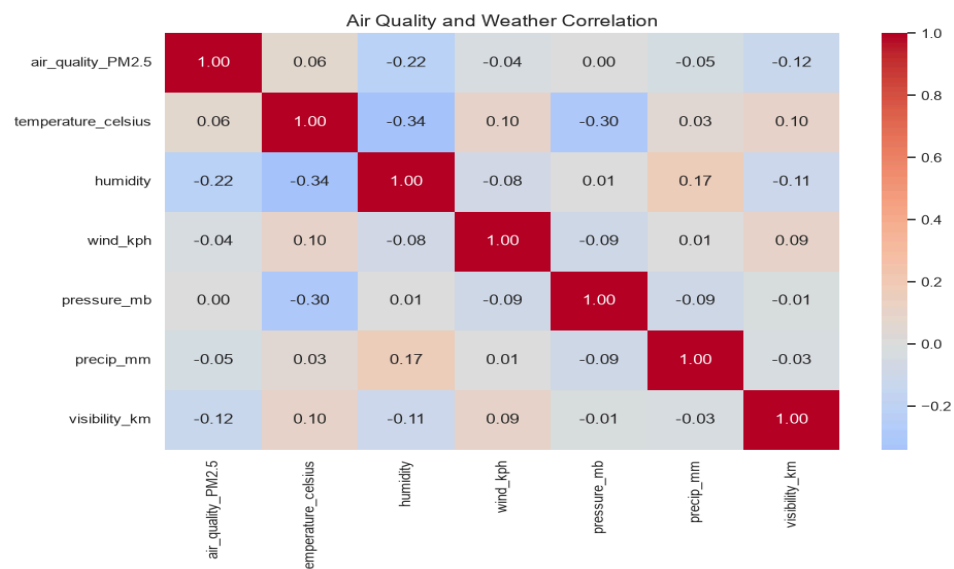
The dataset included several pollutant measurements including: PM2.5, PM10, Carbon monoxide, Nitrogen dioxide, Sulphur dioxide, Ozone, US EPA air-quality index, Great Britain air-quality index.

These variables were analyzed together with weather conditions such as temperature, humidity, precipitation, wind speed, pressure, and visibility.

Correlation Analysis

Correlation analysis was used to examine whether weather conditions were associated with changes in air-quality measurements. The analysis showed that pollutant levels had varying relationships with weather variables. Wind, precipitation,

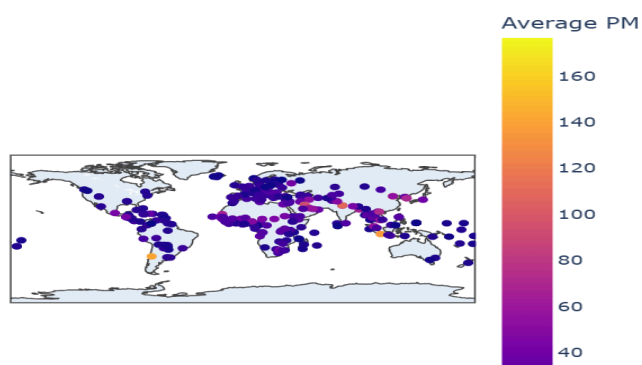
humidity, and visibility were particularly relevant because they can influence how pollutants accumulate, disperse, or are removed from the atmosphere. However, the strength and direction of these relationships differed across pollutants and locations.



PM2.5 Spatial Distribution

PM2.5 values were compared across geographical locations using latitude and longitude. The spatial analysis showed that PM2.5 concentrations were not evenly distributed across the dataset. Some regions recorded higher pollution levels than others, reflecting differences in urban activity, industrial sources, traffic, climate, and local atmospheric conditions.

Global Distribution of Average PM2.5



Interpretation

The results indicate that weather conditions and air quality are related, but the analysis does not establish causation.

For example:

- Stronger wind may help disperse pollutants.
- Rainfall may temporarily reduce airborne particles.
- Low visibility may occur alongside higher particulate concentrations.
- Humidity may affect particle formation and measurement.

These interpretations are consistent with atmospheric behavior, but additional local emissions and meteorological data would be required for stronger conclusions.

Main Findings

- Air-quality levels varied across locations and geographical regions.
- Weather variables showed measurable but generally moderate relationships with pollutant levels.
- PM2.5 displayed clear spatial variation.
- Wind, precipitation, humidity, and visibility were the most relevant environmental variables for interpreting air-quality patterns.
- The findings are correlational and should not be interpreted as proof that weather variables directly caused changes in pollution.

8. FORECASTING METHODOLOGY

Forecasting Location

Forecasting was performed for Bujumbura, Burundi, because it contained the longest suitable daily temperature history in the dataset.

The selected series contained:

Original recorded dates	- 794
Missing calendar dates	- 3
Continuous daily series	- 797
Final modeling observations	- 783

The three missing calendar dates were filled using time-based interpolation to create a continuous daily series.

Forecasting Target

The target variable was: temperature_celsius. The objective was to predict future daily temperature using historical temperature patterns and engineered calendar features.

Forecasting Features - The following features were created:

Feature	Description
lag_1	Temperature one day earlier
lag_7	Temperature seven days earlier

lag_14	Temperature fourteen days earlier
rolling_mean_7	Average temperature over the previous seven days
rolling_std_7	Temperature variability over the previous seven days
month	Calendar month
day_of_year	Position within the year
day_of_week	Day of the week

The first 14 rows were removed because lag_14 was unavailable for those dates.

Chronological Data Split

The forecasting dataset was divided chronologically to preserve the time order and prevent future observations from influencing model training.

Date	Observations	Date Range
Training	548	May 30, 2024 - November 28, 2025
Validation	117	November 29, 2025 - March 25, 2026
Test	118	March 26, 2026 - July 21, 2026

The training set was used to fit the models, the validation set was used to compare model performance and calculate ensemble weights, and the test set was reserved for final evaluation.

Forecasting Models

The following models were evaluated:

Naive Persistence - The naive model predicted that the next day's temperature would be equal to the previous day's temperature. It served as the basic forecasting baseline.

Ridge Regression - Ridge Regression used lag, rolling, and calendar features. The features were standardized using StandardScaler within a pipeline.

SARIMA - SARIMA modeled the ordered temperature series directly using autoregressive, moving-average, and weekly seasonal components. The configuration used was [SARIMA(1,0,1)(1,0,1)_7]

Ensemble Models

Two ensemble approaches were evaluated:

- Simple Average Ensemble
- Validation-Weighted Ensemble

The weighted ensemble assigned higher weights to models with lower validation MSE.

Training and Evaluation Process

The modeling workflow is as follows:

1. Train all models using the training data.
2. Evaluate predictions on the validation period.
3. Calculate ensemble weights using validation MAE.
4. Retrain individual models using the combined training and validation data.
5. Generate predictions for the untouched test period.
6. Compare all individual and ensemble models using the same evaluation metrics.
7. Retrain the best-performing forecasting model using the full historical series.
8. Generate the final seven-day temperature forecast.

The test set was not used for model training, hyperparameter tuning, or ensemble-weight calculation.

Evaluation Metrics

Model performance was evaluated using four metrics:

Mean Absolute Error - MAE measures the average absolute difference between predicted and actual temperatures. Lower values indicate better performance.

Root Mean Squared Error - RMSE gives greater importance to larger prediction errors. Lower values indicate fewer large forecasting mistakes.

R-Squared - R^2 measures how much of the variation in temperature is explained by the model. Values closer to 1 indicate stronger explanatory performance.

Symmetric Mean Absolute Percentage Error - sMAPE measures the average percentage forecasting error while treating overprediction and underprediction symmetrically. Lower values indicate better forecasting accuracy.

9. MODEL RESULTS AND COMPARISON

Validation Results

The models were first trained using the training period and evaluated on the validation period.

Model	MAE (°C)	RMSE (°C)	R ²	sMAPE
Naive Persistence	1.288	1.628	-0.042	5.546 %
Ridge Regression	1.346	1.607	-0.016	5.770 %
SARIMA	1.699	2.012	-0.592	7.260 %
Random Forest	2.641	3.003	-2.546	10.981 %

The naive persistence model achieved the lowest validation MAE and sMAPE, while Ridge Regression achieved the lowest validation RMSE and the R² value closest to zero. These validation results were used to compare model behavior and calculate ensemble weights. The test set was not used during this process.

Ensemble Weights

The validation-weighted ensemble assigned larger weights to models with lower validation MAE.

	Model	Validation MAE	Ensemble Weight
0	Naive Persistence	1.288	0.312
1	Ridge Regression	1.346	0.299
2	SARIMA	1.699	0.237
3	Random Forest	2.641	0.152

The weights summed to 1.0 and were determined entirely from validation performance.

Final Test Results

After validation, the individual models were retrained using the combined training and validation data. All individual and ensemble approaches were then evaluated on the untouched test period.

	Model	Dataset	MAE	RMSE	R ²	sMAPE
0	SARIMA	Test	1.145	1.521	0.083	5.097
1	Simple Average Ensemble	Test	1.158	1.562	0.034	5.150
2	Weighted Ensemble	Test	1.160	1.572	0.021	5.158
3	Ridge Regression	Test	1.195	1.597	-0.010	5.307
4	Random Forest	Test	1.234	1.657	-0.088	5.478
5	Naive Persistence	Test	1.446	1.960	-0.521	6.453

Best Model

SARIMA achieved the strongest final test performance across all four evaluation metrics.

Its test results were:

- MAE: 1.145 °C
- RMSE: 1.521 °C
- R^2 : 0.083
- sMAPE: 5.097 %

An MAE of 1.145°C means that the SARIMA predictions differed from the actual daily temperature by approximately 1.15°C on average.

Compared with the naive persistence baseline, SARIMA reduced:

- MAE by approximately 20.8%
- RMSE by approximately 22.4%
- sMAPE by approximately 21.0%

SARIMA was also the only model to achieve a positive test R^2 score. However, the R^2 value remained modest, indicating that a large proportion of daily temperature variation was not explained by the model.

Ensemble Performance

Both ensemble approaches outperformed the naive persistence baseline but did not outperform standalone SARIMA. The simple-average ensemble slightly outperformed the validation-weighted ensemble on the test period:

- Simple-average ensemble MAE: 1.158 °C
- Validation-weighted ensemble MAE: 1.160 °C

The small difference indicates that validation-based weighting did not provide a meaningful improvement over equal weighting. The ensembles likely performed slightly worse than SARIMA because they combined its predictions with less accurate Ridge Regression, Random Forest, and naive forecasts.

Interpretation

The results demonstrate that a more complex model does not automatically produce better forecasts.

Random Forest performed poorly during validation but improved after retraining with additional historical observations. Ridge Regression produced competitive results but did not clearly outperform SARIMA or the ensemble methods.

SARIMA was the most effective approach for the selected location and test period because it modeled the ordered temperature series and repeating weekly patterns directly.

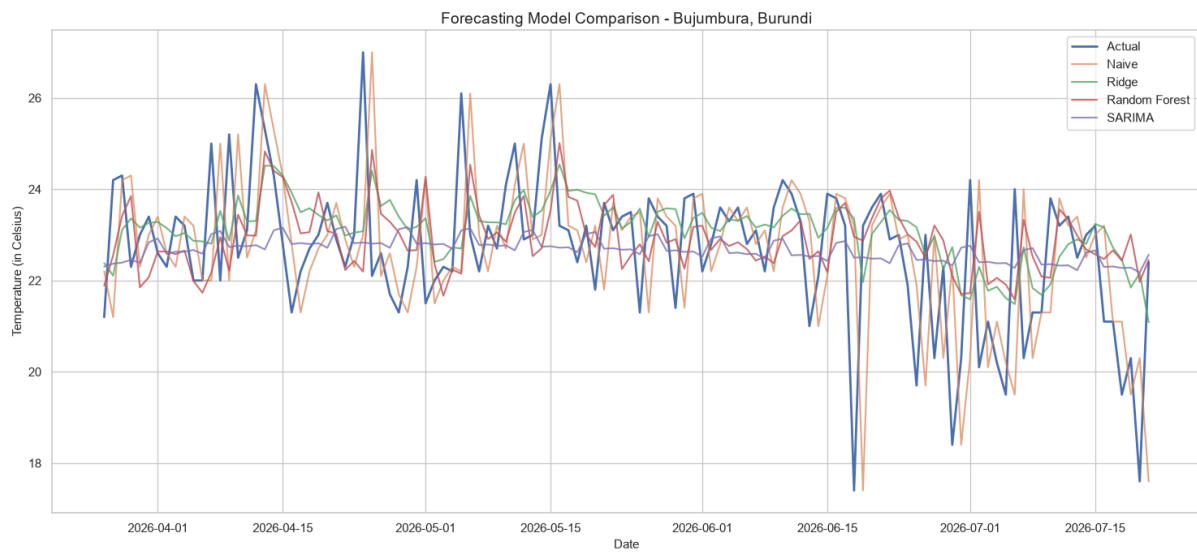


Figure: Final Test MAE comparison across all forecasting models

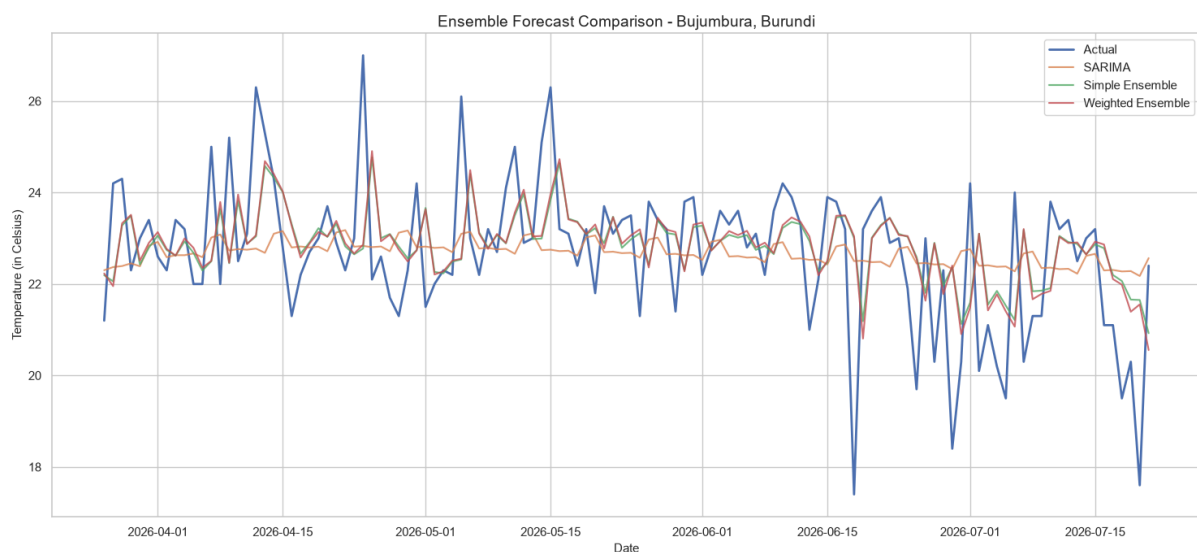


Figure: Actual versus predicted test temperatures for SARIMA and the ensemble models.

10. FEATURE-IMPORTANCE ANALYSIS

Methods

Feature importance was evaluated for the machine-learning forecasting features using three methods:

- Correlation with the target temperature
- Random Forest built-in feature importance
- Permutation importance on the test set

SARIMA was not included because it models the temperature series directly and does not use the engineered feature matrix.

Correlation-Based Importance

Correlation analysis measured the linear relationship between each feature and the current temperature.

	Feature	Correlation with Temperature
0	rolling_mean_7	0.646
1	lag_1	0.562
2	lag_7	0.448
3	lag_14	0.407
4	day_of_year	0.193
5	month	0.190
6	rolling_std_7	0.170
7	day_of_week	-0.012

The seven-day rolling mean had the strongest correlation with the target temperature. Recent lagged temperatures were also more strongly related to the target than calendar features.

Random Forest Importance

Random Forest importance measured how strongly each feature contributed to decision-tree splits.

	Feature	Random Forest Importance
0	rolling_mean_7	0.338
1	lag_1	0.230
2	rolling_std_7	0.123
3	lag_14	0.090
4	lag_7	0.083
5	day_of_year	0.082
6	day_of_week	0.040
7	month	0.014

The Random Forest model also ranked rolling_mean_7 as the most influential feature, followed by lag_1.

Permutation Importance

Permutation importance measured how much forecasting performance changed when each feature was randomly shuffled.

	Feature	Permutation Importance	Importance Standard Deviation
0	lag_1	0.043	0.028
1	rolling_mean_7	0.033	0.016
2	lag_7	0.010	0.016
3	day_of_year	0.009	0.008
4	lag_14	0.005	0.008
5	month	0.002	0.002
6	day_of_week	0.001	0.007
7	rolling_std_7	-0.016	0.022

lag_1 produced the highest permutation importance, indicating that the previous day's temperature had the greatest effect on Random Forest test performance. The negative value for rolling_std_7 indicates that shuffling this feature did not reduce performance and may have slightly improved it. This suggests that the feature provided limited predictive value during the test period.

Comparison of Methods

	Method	Most Important Feature
0	Correlation	rolling_mean_7
1	Random Forest	rolling_mean_7
2	Permutation Importance	lag_1

The methods produced slightly different rankings because they measure feature importance differently. However, all three showed that recent temperature history was more useful than calendar information.

Main Findings

- rolling_mean_7 was the most important feature according to correlation and Random Forest importance.
- lag_1 was the most important feature according to permutation importance.
- lag_7 and lag_14 provided additional historical information but were less influential.
- Calendar features such as month, day_of_year, and day_of_week had relatively low importance.
- day_of_week had almost no meaningful relationship with temperature.
- Recent temperature values were therefore the strongest predictors used by the machine-learning models.

11. SEVEN DAY TEMPERATURE FORECAST

Final Model Selection

SARIMA was selected for the final forecast because it achieved the best performance on the test dataset.

Its test performance was:

- MAE: 1.145 °C
- RMSE: 1.521 °C
- R²: 0.083
- sMAPE: 5.097%

After model evaluation was completed, SARIMA was retrained using all 797 daily temperature observations available for Bujumbura, Burundi.

The complete historical period extended from May 16, 2024 through July 21, 2026.

Seven-Day Forecast Results

The final model generated temperature predictions for the seven calendar days immediately following the latest observation in the dataset.

	Forecast Date	Predicted Temperature (in Celsius)	Lower 95% Interval (in Celsius)	Upper 95% Interval (in Celsius)
0	2026-07-22	20.940	16.380	25.500
1	2026-07-23	21.140	16.520	25.750
2	2026-07-24	21.000	16.330	25.670
3	2026-07-25	21.130	16.400	25.850
4	2026-07-26	21.060	16.280	25.840
5	2026-07-27	21.180	16.350	26.010
6	2026-07-28	20.950	16.060	25.830

Forecast Interpretation

The predicted temperatures remained stable across the seven-day period. The main forecast statistics were:

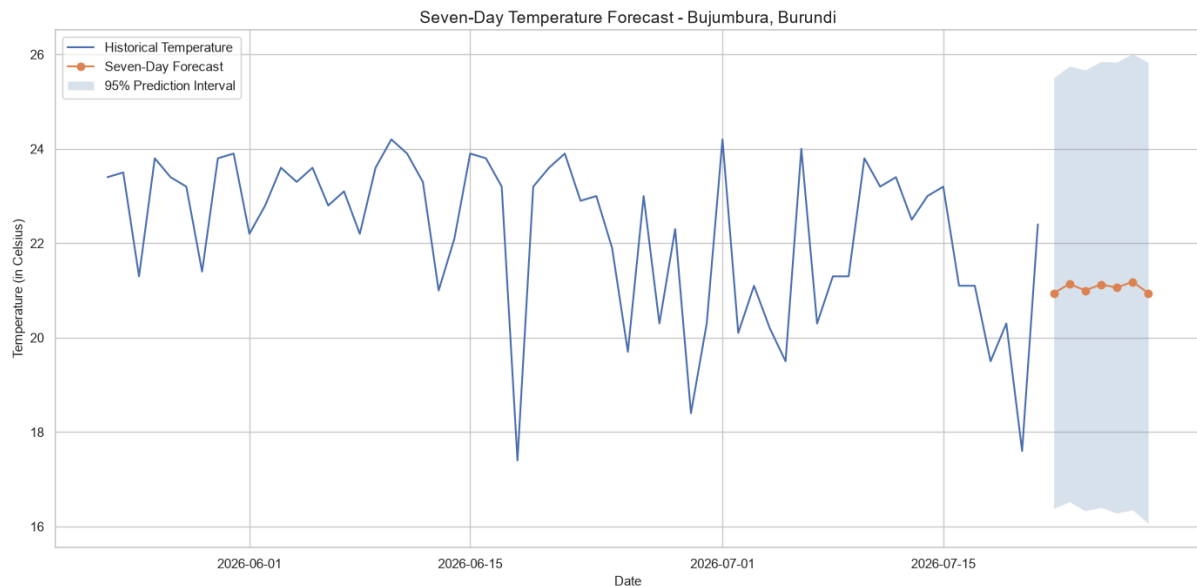
- Minimum predicted temperature: 20.94°C
- Maximum predicted temperature: 21.18°C
- Average predicted temperature: approximately 21.06°C
- Overall forecast range: 0.24°C

The small variation between daily point forecasts indicates that SARIMA expected relatively stable temperature conditions during the forecast period. The 95% prediction intervals were much wider than the point forecasts, generally ranging from approximately 16°C to 26°C. These intervals represent forecast uncertainty and show that actual daily temperatures could vary more substantially than the central predictions suggest. The uncertainty intervals gradually widened across the forecast horizon because predictions farther into the future generally contain greater uncertainty.

Forecast Scope

The forecast was generated using historical daily temperature patterns only. It didn't include future observations or forecasts for: Humidity, Atmospheric pressure,

Precipitation, Wind speed, Cloud cover, Satellite conditions, Large-scale climate systems. Therefore, the forecast demonstrates the time-series modeling workflow and should not be interpreted as a replacement for an operational meteorological forecast.



12. LIMITATIONS AND FUTURE IMPROVEMENTS

Limitations

- The dataset covered a limited historical period, so it was not sufficient for reliable long-term climate-change conclusions.
- Forecasting was performed only for Bujumbura, Burundi. Model performance may differ for other cities and climate regions.
- Three missing calendar dates were filled using interpolation, meaning those values were estimated rather than directly observed.
- The forecasting models relied mainly on historical temperature and calendar features.
- Future humidity, wind, pressure, precipitation, cloud cover, and satellite information were not available to the models.
- External geographical factors such as elevation, terrain, and coastal proximity were not included.
- SARIMA achieved the best performance, but its R^2 score of 0.083 was modest, showing that much of the daily temperature variation remained unexplained.
- Ridge Regression and Random Forest used lag features created from observed temperatures, while SARIMA generated a direct multi-step forecast. Therefore, the evaluation approaches were not perfectly identical.
- Isolation Forest identified statistically unusual observations but could not determine whether they represented real extreme weather or measurement errors.

- Air-quality relationships were correlational and did not establish causation.
- The final seven-day forecast was intended to demonstrate the forecasting workflow and should not replace an operational weather service.

Future Improvements

Future work could improve the project by:

- Applying the forecasting workflow to multiple cities and climate regions.
- Using a longer historical dataset to capture stronger seasonal patterns.
- Performing rolling time-series cross-validation.
- Tuning SARIMA parameters using validation-based grid search.
- Including lagged humidity, pressure, wind, precipitation, and cloud-cover features.
- Adding elevation, terrain, ocean, and satellite information.
- Testing additional models such as XGBoost, LightGBM, Prophet, or recurrent neural networks.
- Using a consistent rolling one-step-ahead evaluation approach for all models.
- Improving uncertainty estimation through probabilistic forecasting methods.
- Developing an interactive dashboard for weather trends, anomalies, air quality, and forecasts.

13. CONCLUSION

This project analyzed global weather, geographical, and air-quality patterns and compared multiple temperature-forecasting approaches using the Global Weather Repository dataset.

For forecasting, Bujumbura, Burundi, was selected because it provided the longest suitable daily temperature history. Naive Persistence, Ridge Regression, Random Forest, SARIMA, and two ensemble approaches were compared using a chronological training, validation, and test split.

SARIMA achieved the best test performance with an MAE of 1.145°C, outperforming the naive baseline by approximately 20.8%. Feature-importance analysis showed that recent temperature history, particularly the seven-day rolling average and previous day's temperature, was more useful than calendar features.

The final SARIMA model generated a seven-day forecast ranging from 20.94°C to 21.18°C. Overall, the project satisfied the assessment requirements and demonstrated a complete workflow from data preparation and analysis to model evaluation and future forecasting.

14. PM ACCELERATOR MISSION

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